

ISRC Robotics Event: Line Following Event Rules, Regulations, and Judging Criteria

Rules and Regulations

1. Eligibility

- ❑ **Age Group:** This event is available for participants aged 19-22 only.
- ❑ **Team Size:** Teams must consist of 2-5 members.

2. Event Structure

- ❑ **Rounds:**
 - o **First Round:** All teams will participate in the first round.
 - o **Second Round:** Teams that pass the first round will compete in the second round.
 - o **Final Round:** The top-performing teams from the second round will advance to the final round.
- ❑ **Attempts:** In each round, teams will be given two attempts to complete the track. The best attempt (shortest time with the least penalties) will be considered for scoring.

3. Track Specifications

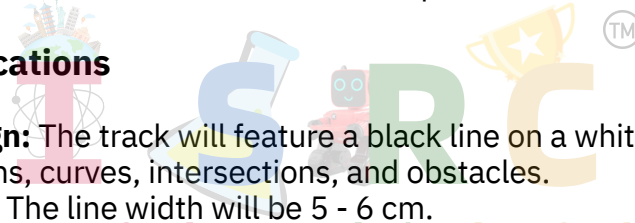
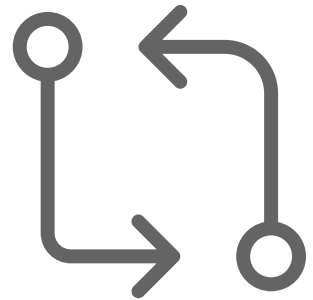
- ❑ **Track Design:** The track will feature a black line on a white surface, including straight paths, curves, intersections, and obstacles.
- ❑ **Line Width:** The line width will be 5 - 6 cm.
- ❑ **Track Length:** The track will range from 15 to 25 meters in length, depending on the complexity.
- ❑ **Obstacles:** The track will include challenges such as bumps, tunnels, and ramps to test the robot's navigation and adaptability.

4. Robot Requirements

- ❑ **Autonomous Operation:** Robots must operate autonomously with no external control during the event.
- ❑ **Sensor Usage:** Teams are encouraged to use sensors like IR sensors, light sensors, or cameras to detect and follow the line.
- ❑ **Power Supply:** Robots must be powered by batteries. Teams should have spare batteries on hand.

5. Competition Format

- ❑ **Time Trials:** The robot's performance is timed, and the fastest time to complete the track with the fewest penalties is recorded.
- ❑ **Obstacle Navigation:** Robots must navigate through obstacles without leaving the track. Deviating from the track or getting stuck will incur penalties.



- ❑ **Restarts:** Teams are allowed to restart their robot if it leaves the track, but each restart will incur a time penalty.

6. Scoring and Penalties

- ❑ **Time Scoring:** The robot with the shortest time and fewest penalties in each round will receive the highest score.
- ❑ **Penalty Points:**
 - Leaving the track: +10 seconds
 - Colliding with an obstacle: +5 seconds
 - Restarting the robot: +15 seconds
- ❑ **Final Scoring:** The best score from the two attempts in each round will be considered for advancing to the next round and for final scoring.

7. Safety Regulations

- ❑ **Robot Safety:** Robots must be designed safely with no sharp edges or hazardous materials.
- ❑ **Track Safety:** The track area must be kept clear of all participants and spectators during the runs.
- ❑ **Emergency Stop:** Each robot must have an accessible emergency stop mechanism.



Judging Criteria for Line Following Event

Scoring Out of 1000 Points:

1. Innovation and Originality (200 points)

- ❑ **Creative Navigation Techniques (80 points):** Evaluate the originality of the robot's design and its approach to line following, including the use of innovative sensors or algorithms.
- ❑ **Advanced Problem-Solving (60 points):** Assess the robot's ability to handle unexpected challenges or deviations in the track.
- ❑ **Novelty in Design (60 points):** Consider the uniqueness of the robot's physical design and its potential to introduce new concepts in robotics.

2. Technical Mastery (250 points)

- ❑ **Precision in Line Following (100 points):** Evaluate the accuracy with which the robot follows the line, maintaining consistent alignment and speed.
- ❑ **Sensor Integration (80 points):** Assess the effectiveness of the sensors used for line detection, including their calibration and response to track variations.

- ❑ **Robustness and Durability (70 points):** Evaluate the robot's ability to function reliably under various conditions, including the presence of obstacles or changes in track surface.

3. Practical Applicability and Global Impact (150 points)

- ❑ **Real-World Applications (70 points):** Consider the potential of the robot's design to be applied in real-world scenarios, such as automated vehicles or industrial automation.
- ❑ **Environmental Considerations (50 points):** Evaluate the sustainability of the robot, including energy efficiency and material usage.
- ❑ **Scalability (30 points):** Assess the potential for the robot's design to be scaled for larger or more complex applications.

4. Presentation Excellence (200 points)

- ❑ **Technical Explanation (100 points):** Evaluate the team's ability to explain the technical aspects of their robot, including sensor integration, programming, and design choices.
- ❑ **Demonstration Quality (70 points):** Assess the effectiveness of the live demonstration, including the robot's performance and the clarity of the team's presentation.
- ❑ **Team Communication (30 points):** Consider how well the team collaborates and communicates their ideas during the presentation.

5. Judges' Challenge (100 points)

- ❑ **Real-Time Problem Solving (50 points):** Present a real-time challenge to the teams, such as modifying the track or introducing a new obstacle, and evaluate their ability to adapt.
- ❑ **Cross-Disciplinary Collaboration (30 points):** Assess how well the team integrates knowledge from different STEM fields, such as engineering, programming, and design.
- ❑ **Time-Constrained Innovation (20 points):** Evaluate the team's ability to innovate under time constraints during the challenge.

6. Robot Design and Functionality (100 points)

- ❑ **Mechanical Design (40 points):** Assess the structural design, including the materials used, compactness, and overall build quality of the robot.
- ❑ **Programming and Control (40 points):** Evaluate the robot's programming logic, control algorithms, and efficiency in executing tasks.
- ❑ **Adaptability (20 points):** Consider the robot's ability to adapt to different track conditions, including speed adjustments and obstacle avoidance.